

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CBS) (CBCS) Degree Examinations

December 2024 – January 2025

22CBS102 – CYBER FORENSICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What is digital forensics? Explain the digital forensics evidence collection and processing framework	7	L*3	1	1
	b)	Define cyber crime. Explain the different types of cyber crimes.	7	L2	1	1
	c)	Differentiate between FAT and NTFS file system.	6	L2	1	1
2.	a)	List and Explain the Five hives of windows registry in detail.	7	L2	1	1
	b)	Explain the auditing and security architecture in cyber forensics.	7	L3	1	1
	c)	Outline the applications of digital forensics.	6	L2	1	1
		Unit – II				
3.	a)	Explain the commands involved in Linux OS for collecting the information from the system.	7	L2	2	1
	b)	List and explain the several different pieces of Linux.	7	L2	2	1
	c)	Explain the scenario of collecting the volatile data through different command line arguments.	6	L2	2	1
4.	a)	What is device driver? Illustrate the Linux device driver architecture with suitable diagram.	7	L3	2	1
	b)	Explain the Filesystem Hierarchy Structure (FHS) and also list different types of Linux log files in Linux OS.	7	L2	2	1
	c)	Differentiate between EXT 2/3/4 file systems.	6	L2	2	1
		Unit – III				
5.	a)	Explain the steps involved in performing the forensics on database.	7	L2	3	1
	b)	What are Transaction Logs? Explain the key functions and contents of Transaction logs.	7	L2	3	1
	c)	What are the various indexing used in Table storage? Explain.	6	L2	3	1
6.	a)	Illustrate the steps involved in recovering the database if it is deleted.	7	L3	3	1
	b)	What is database tampering? What are various ways to tamper the database and how to prevent it.	7	L2	3	1
	c)	Explain the features of two SQL Server Forensics Tool.	6	L2	3	1
		Unit – IV				
7.	a)	With suitable diagram explain the functionalities of each layer of OSI model.	7	L2	4	1
	b)	With appropriate diagram explain the Image Forensic analysis Process.	7	L2	4	1
	c)	List and explain the command lines to filter on IP address, network and MAC address.	6	L2	4	1
8.	a)	What is Social Media forensics? Explain the various stages and tools applicable with each stages in the process of Social media forensics.	7	L3	4	1
	b)	Why there is a need to capture the packets? Explain any 2 packet capturing tools in detail.	7	L2	4	1
	c)	Discuss the Email investigation process with suitable illustration.	6	L2	4	1

Unit – V

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| 9. | a) | Discuss the different types of offenses and sections under ITA act 2000. | 10 | L2 | 5 | 1 |
| | b) | What are the skills required by the Cyber forensics expert and also how data mining is used in forensic? Explain with suitable illustration. | 10 | L3 | 5 | 1 |
| 10. | a) | Discuss any two case study with respect to Cybercrime. | 10 | L2 | 5 | 1 |
| | b) | What is DOS attack? Explain the different types or levels of DOS attack and also list the tools used to launch the DOS attack. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
